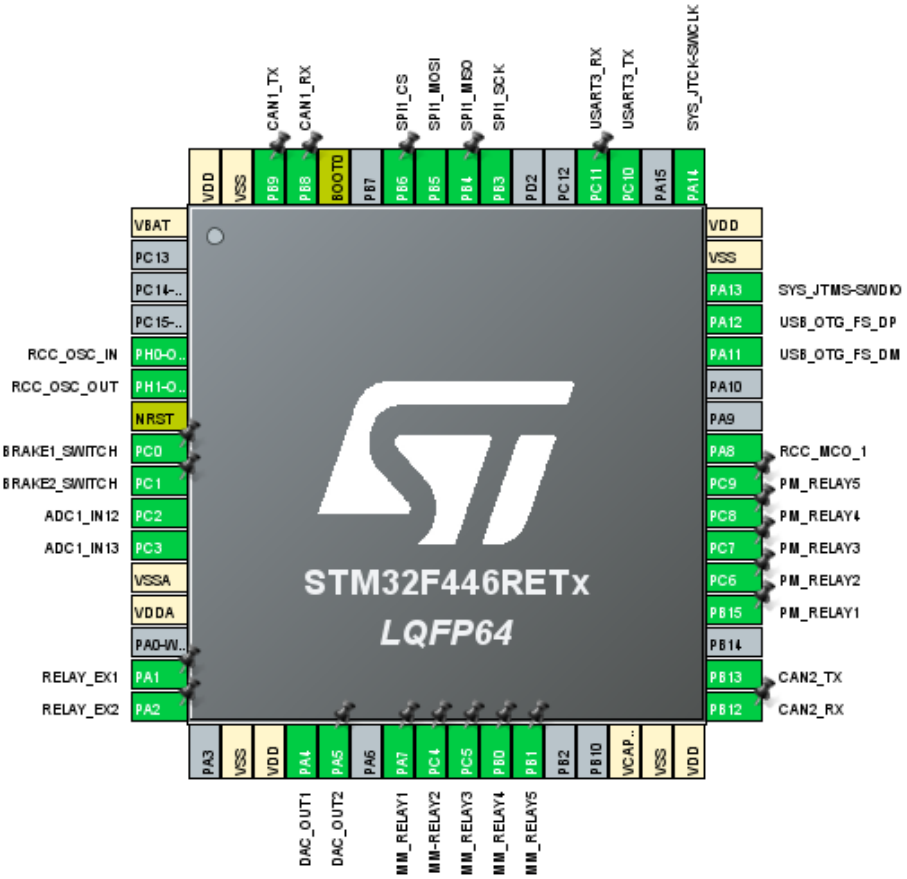


STM32 Assigned Pins:

Name	Type	Signal	Label
PH0-OSC_IN	I/O	RCC_OSC_IN	
PH1-OSC_OUT	I/O	RCC_OSC_OUT	
PC0	Input	GPIO_Input	BRAKE1_SWITCH
PC1	Input	GPIO_Input	BRAKE2_SWITCH
PC2	I/O	ADC1_IN12	
PC3	I/O	ADC1_IN13	
PA1	Output	GPIO_Output	RELAY_EX1
PA2	Output	GPIO_Output	RELAY_EX2
PA4	I/O	DAC_OUT1	
PA5	I/O	DAC_OUT2	
PA7	Output	GPIO_Output	MM_RELAY1
PC4	Output	GPIO_Output	MM-RELAY2
PC5	Output	GPIO_Output	MM_RELAY3
PB0	Output	GPIO_Output	MM_RELAY4
PB1	Output	GPIO_Output	MM_RELAY5
PB12	I/O	CAN2_RX	
PB13	I/O	CAN2_TX	

PB15	Output	GPIO_Output	PM_RELAY1
PC6	Output	GPIO_Output	PM_RELAY2
PC7	Output	GPIO_Output	PM_RELAY3
PC8	Output	GPIO_Output	PM_RELAY4
PC9	Output	GPIO_Output	PM_RELAY5
PA8	I/O	RCC_MCO_1	
PA11	I/O	USB_OTG_FS_DM	
PA12	I/O	USB_OTG_FS_DP	
PA13	I/O	SYS_JTMS-SWDIO	
PA14	I/O	SYS_JTCK-SWCLK	
PC10	I/O	USART3_TX	
PC11	I/O	USART3_RX	
PB3	I/O	SPI1_SCK	
PB4	I/O	SPI1_MISO	
PB5	I/O	SPI1_MOSI	
PB6	Output	GPIO_Output	SPI1_CS
PB8	I/O	CAN1_RX	
PB9	I/O	CAN1_TX	

Hardware layout and configurations:

A. Connector

Connector	Pin		STM32 pin		Function	
J1 USB B-mini	1		-		VBUS	
	2		PA12		USB D+	
	3		PA11		USB D-	
	4		-		ID (not connected)	
	5		-		GND	
	6		-		Shield (connected to GND)	
J2 ST-LINK V2	1		-		3v3	
	2		PA13		SWDIO	
	3		PA14		SWCLK	
	4		NRST		NRST	
	5		-		GND	
J3 Left STM32 Extension Pins	1	2	PC13	-	PC13	3V3
	3	4	PC14	-	PC14	3V3
	5	6	PC15	-	PC15	3V3
	7	8	NRST	-	NRST	3V3
	9	10	PC0	-	PC0	3V3
	11	12	PC1	-	PC1	3V3
	13	14	PC2	-	PC2	GND
	15	16	PC3	-	PC3	GND
	17	18	PA0	-	PA0	GND
	19	20	PA1	-	PA1	GND
	21	22	PA2	-	PA2	GND
	23	24	PA3	-	PA3	GND

	25	26	PA4	-	PA4	GND
J4 Bottom STM32 Extension Pins	1	2	PA5	-	PA5	5V
	3	4	PA6	-	PA6	5V
	5	6	PA7	-	PA7	5V
	7	8	PC4	-	PC4	5V
	9	10	PC5	-	PC5	5V
	11	12	PB0	-	PB0	5V
	13	14	PB1	-	PB1	GND
	15	16	PB2	-	PB2	GND
	17	18	PB10	-	PB10	GND
	19	20	PB12	-	PB12	GND
	21	22	PB13	-	PB13	GND
	23	24	PB14	-	PB14	GND
J5 Right STM32 Extension Pins	1	2	PB15	-	PB15	5V
	3	4	PC6	-	PC6	5V
	5	6	PC7	-	PC7	5V
	7	8	PC8	-	PC8	5V
	9	10	PC9	-	PC9	5V
	11	12	PA8	-	PA8	5V
	13	14	PA9	-	PA9	GND
	15	16	PA10	-	PA10	GND
	17	18	PA11	-	PA11	GND
	19	20	PA12	-	PA12	GND
	21	22	PA13	-	PA13	GND
	23	24	PA14	-	PA14	GND
J6 Top STM32 Extension Pins	1	2	PA15	-	PA15	3V3
	3	4	PC10	-	PC10	3V3
	5	6	PC11	-	PC11	3V3
	7	8	PC12	-	PC12	3V3
	9	10	PD2	-	PD2	3V3
	11	12	PB3	-	PB3	3V3
	13	14	PB4	-	PB4	GND
	15	16	PB5	-	PB5	GND
	17	18	PB6	-	PB6	GND
	19	20	PB7	-	PB7	GND
	21	22	PB8	-	PB8	GND
	23	24	PB9	-	PB9	GND
J7 RELAY1_EXTERNAL	1		-		External relay 1 NO pin	
	2		-		External relay 1 COM pin	
J8 RELAY2_EXTERNAL	1		-		External relay 2 NO pin	
	2		-		External relay 2 COM pin	
J9	1	9	-	-	MAIN_DGND	MAIN_DGND
	2	10	-	-	N/C	MAIN_TGND

MAIN MOTOR	3	11	-	-	DRIVER_MAIN1	DRIVER_MAIN2
	4	12	-	-	CAN1_P	DRIVER_MAIN3
	5	13	-	-	CAN1_N	DRIVER_MAIN4
	6	14	-	-	DRIVER_MAIN5	MAIN_THROTTLE
	7	15	-	-	N/C	MAIN_DRIVER
	8	-	-	-	MAIN_DRIVER	-
J10 PUMP MOTOR	1	2	-	-	PUMP_DGND	PUMP_DGND
	3	4	-	-	N/C	PUMP_TGND
	5	6	-	-	DRIVER_PUMP1	DRIVER_PUMP2
	7	8	-	-	CAN2_P	DRIVER_PUMP3
	9	10	-	-	CAN2_N	DRIVER_PUMP4
	11	12	-	-	DRIVER_PUMP5	PUMP_THROTTLE
	13	14	-	-	N/C	PUMP_DRIVER
	15	-	-	-	PUMP_DRIVER	-
J11 BRAKE1	1		-		BRAKE1_SWITCH	
	2		-		GND	
J12 BRAKE2	1		-		BRAKE2_SWITCH	
	2		-		GND	
J13 CN_MCP2551 (CAN1)	1		-		CAN1_N	
	2		-		CAN1_P	
	3		PB8		CAN1_RX	
	4		PB9		CAN1_TX	
	5		-		GND	
	6		-		5V	
J14 CN_MCP2551 (CAN2)	1		-		CAN2_N	
	2		-		CAN2_P	
	3		PB12		CAN2_RX	
	4		PB13		CAN2_TX	
	5		-		GND	
	6		-		5V	
J15 CAN1 TEST	1		-		CAN1_N	
	2		-		CAN1_P	
J16 CAN2 TEST	1		-		CAN2_N	
	2		-		CAN2_P	

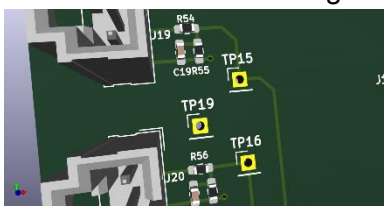
J17 SD CARD MODULE	1	-	N/C
	2	PB6	SPI1_CS
	3	PB3	SPI1_SCK
	4	PB4	SPI1_MOSI
	5	PB5	SPI1_MISO
	6	-	5V
	7	-	GND
J18 USART	1	-	GND
	2	PC10	USART3_TX
	3	PC11	USART3_RX
J19 ADC1_IN12	1	-	5V
	2	-	SIGNAL
	3	-	GND
J20 ADC1_IN13	1	-	5V
	2	-	SIGNAL
	3	-	GND
J21 Power Supply	1	-	12V INPUT
	2	-	GND
	3	-	GND
	4	-	72V INPUT

1st pin indicators:

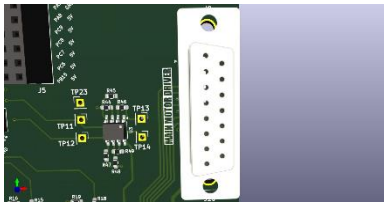
- Pin headers: Right angle mark at the side of the connector.



- Molex connectors: Right angle mark at the side of connector.



- D-SUB connectors: White triangle towards the first pin of the connector.

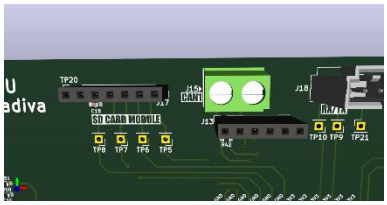


Pinout pattern for D-SUB connector:

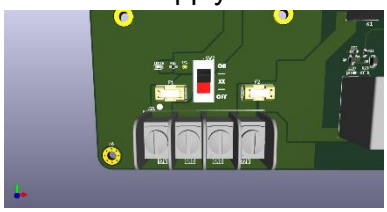
8 — 7 — 6 — 5 — 4 — 3 — 2 — 1

15—14—13—12—11—10—9

- Small terminal block: White triangle mark next to the connector.



- Power supply terminal block: White circle towards the first pin of the connector.



B. Jumper

Jumper	Pin	STM32 pin	Function
JP1 BOOT0MODE	1	-	3V3
	2	BOOT0	BOOT0
	3	-	GND
JP2 BOOT1MODE	1	-	3V3
	2	BOOT1/PB2	BOOT1
	3	-	GND

JP1:

- Connect pin 1-2: Pull BOOT0 up → The logic state of BOOT0 is 1
- Connect pin 2-3: Pull BOOT0 down → The logic state of BOOT0 is 0

JP2:

- Connect pin 1-2: Pull BOOT1 up → The logic state of BOOT1 is 1
- Connect pin 2-3: Pull BOOT1 down → The logic state of BOOT1 is 0

Boot modes in STM32:

Boot Mode Pins		Boot Mode
BOOT1	BOOT0	
X (don't care)	0	Flash memory
0	1	System memory
1	1	Embedded SRAM

To use ST-LINK V2 for debugging, set BOOT0 to logic 0 and BOOT1 to any logic state (do not put it floating)

C. Switch

S1 – Reset switch

S2 – Power switch

D. LED

LED1 – VBUS power

LED2 – Main motor relay 1

LED3 – Main motor relay 2

LED4 – Main motor relay 3

LED5 – Main motor relay 4

LED6 – Main motor relay 5

LED7 – Pump motor relay 1

LED8 – Pump motor relay 2

LED9 – Pump motor relay 3

LED10 – Pump motor relay 4
LED11 – Pump motor relay 5
LED12 – External relay 1
LED13 – External relay 2
LED14 – 5V power
LED15 – 3V3 power
LED16 – 12V power

E. Test Point

TP1 – 12V power
TP2 – 5V power
TP3 – 3V3 power
TP4 – MCO Microcontroller Clock Output (PA8)
TP5 – SPI1_CS
TP6 – SPI1_SCK
TP7 – SPI1_MOSI
TP8 – SPI1_MISO
TP9 – USART3_TX
TP10 – USART3_RX
TP11 – DAC1_OUT
TP12 – DAC2_OUT
TP13 – MAIN_THROTTLE
TP14 – PUMP_THROTTLE
TP15 – ADC1_IN12
TP16 – ADC1_IN13
TP17 – BRAKE1_IN
TP18 – BRAKE2_ IN
TP19, TP20, TP21, TP22, TP23 – Ground point for test points (used to give a closer ground point to an area of test points)

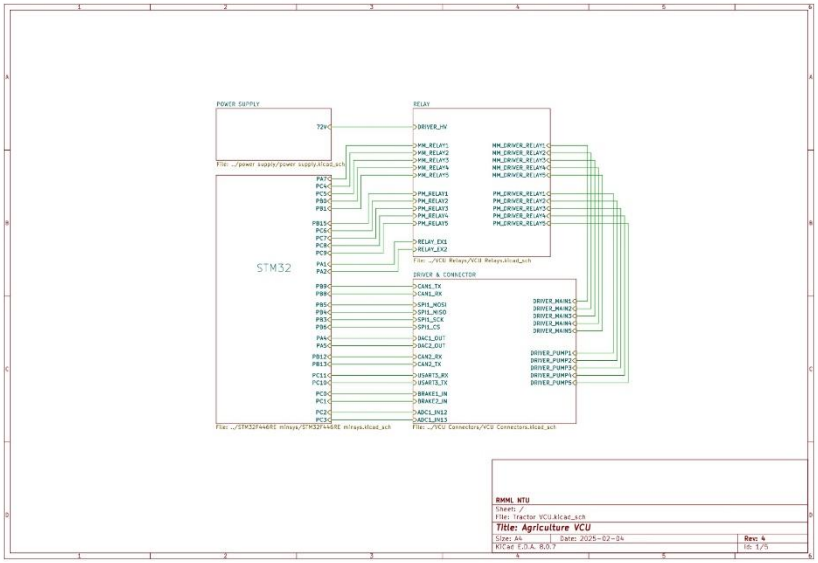
F. PCB Standoff

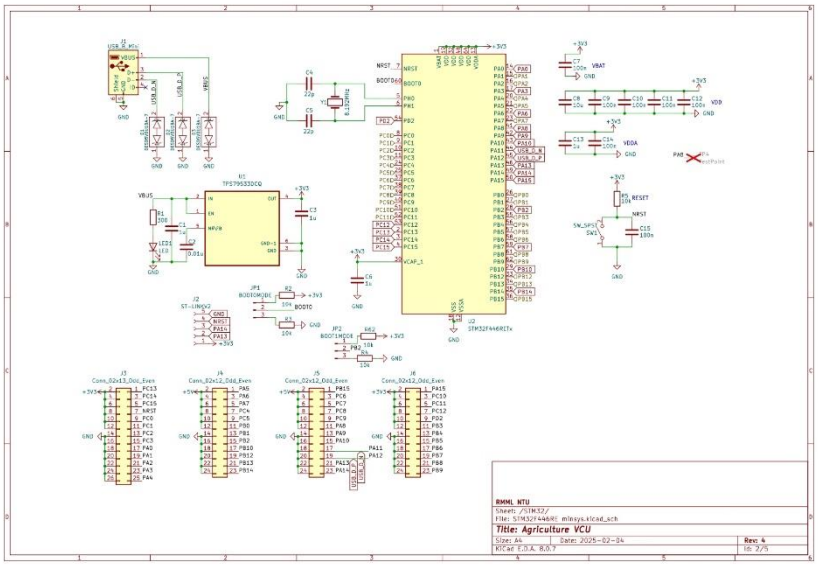
MP1 – GND
MP2 – BUCK INPUT
MP3 – BUCK OUTPUT
MP4 – GND

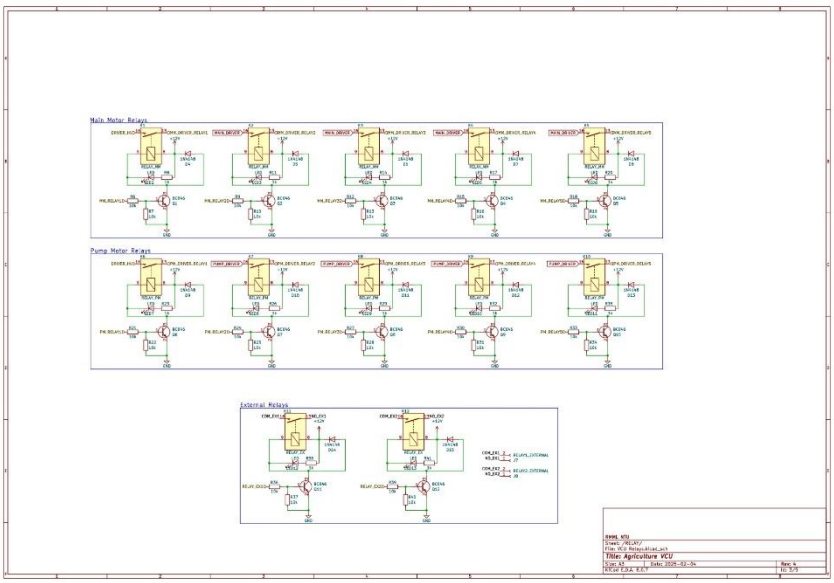
G. Fuse

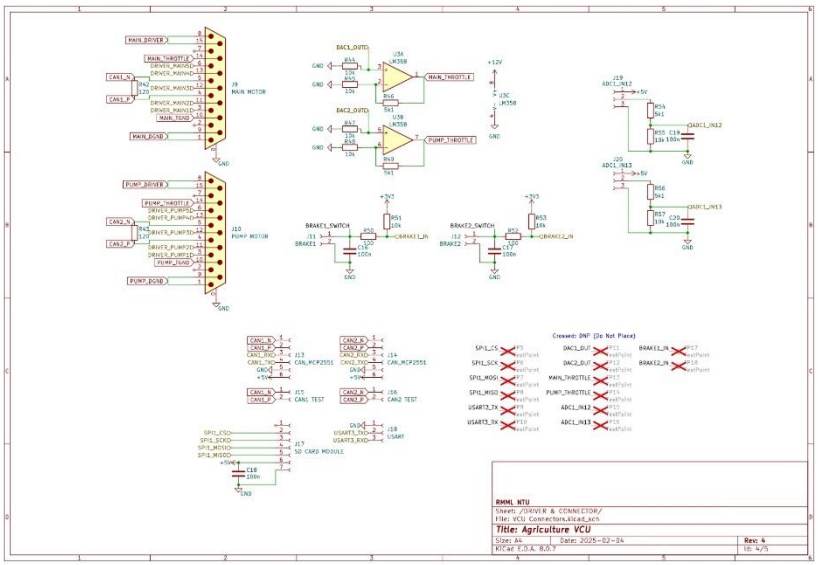
F1 – 12V input
F2 – 72V

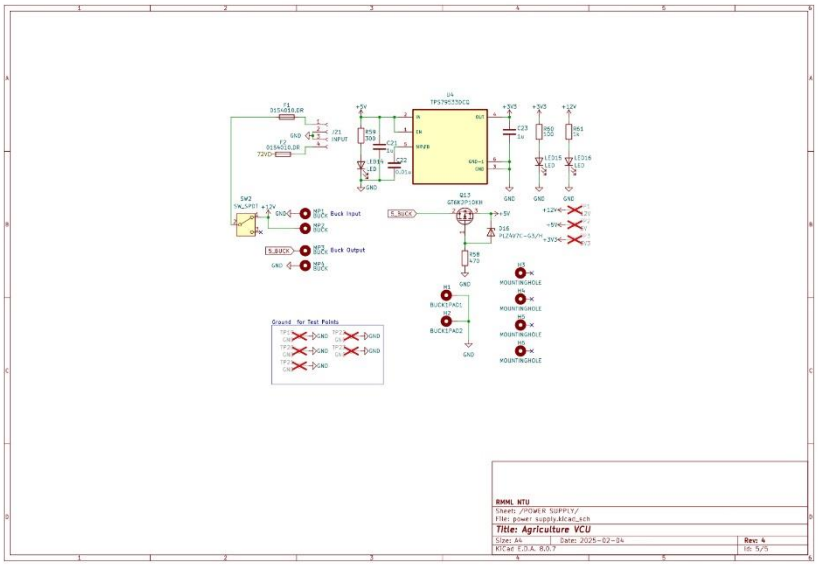
H. Schematic


POWER SUPPLY
RELAY
STM32
DRIVER & CONNECTOR
RMML NTU
Title: Agriculture VCU


RMML NTU
Title: Agriculture VCU


Main Motor Relays
Pump Motor Relays
External Relays
RMML NTU
Title: Agriculture VCU


MAIN_DRIVER
MAIN_THROTTLE
CAN1_N
CAN1_P
MAIN_5GND
J9
MAIN MOTOR
PUMP_DRIVER
PUMP_THROTTLE
CAN2_N
CAN2_P
PUMP_5GND
J10
PUMP MOTOR
DAC1_OUT0
DAC2_OUT0
U3A
U3B
U3C
MCP6N
MAIN_THROTTLE
PUMP_THROTTLE
+12V
GND
J19
ADC1_IN12
J20
ADC1_IN13
BRAKE1_SWITCH
BRAKE2_SWITCH
BRAKE1
BRAKE2
BRAKE1_IN
BRAKE2_IN
+3V3
J13
CAN_MCP2551
J14
J15
CAN1 TEST
J16
CAN2 TEST
J17
SD CARD MODULE
J18
USART3
Crossed: DNP (Do Not Place)
SPI1_CS
SPI1_SCK
SPI1_MOSI
SPI1_MISO
USART3_TX
USART3_RX
DAC1_OUT
DAC2_OUT
MAIN_THROTTLE
PUMP_THROTTLE
ADC1_IN12
ADC1_IN13
BRAKE1_IN
BRAKE2_IN
BMML NTU
Sheets: /DRIVER & CONNECTOR/
File: VCU Connectors.kicad_sch
Title: Agriculture VCU
Size: A4
Date: 2025-02-24
Rev: 4
KiCad F.O.A. 8.0.7
Id: 4/5


F1
D15H10.0R
F2
72VDC
J21
INPUT
U4
TPS7H5310-Q1
+2V
+3V3
+12V
SW2
SW_SPDT
S_BUCK
MP1
Buck Input
MP2
Buck Output
Q13
GTK03104H
D16
PLZ4V7C-G3/H
LED15
LED16
H1
BUCK1PAD1
H2
BUCK1PAD2
H3
MOUNTINGHOLE
H4
MOUNTINGHOLE
H5
MOUNTINGHOLE
H6
MOUNTINGHOLE
Ground for Test Points
BMML NTU
Sheets: /POWER SUPPLY/
File: power supply.kicad_sch
Title: Agriculture VCU
Size: A4
Date: 2025-02-24
Rev: 4
KiCad F.O.A. 8.0.7
Id: 5/5